varsha elango

PROJECTS

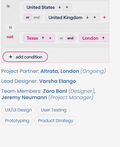
Is United States, Not Texas.

Sharing Codes

The Yellow Juice

Is United States, Not Texas.

Design boolean logic conditions (AND/OR/NOT) into the Advanced search filters of a people intelligence platform.



Project Partner: **Altrata, London** (Ongoing)
Lead Designer: **Varsha Elango**
Team Members: **Zara Bani** (Designer), **Jeremy Heumann** (Project Manager)

UX/UI Design | User Testing | Prototyping | Product Strategy

The problem: Users couldn't search for "CEOs in the US but not Texas, who donated $10k to education but not to Harvard." AND/OR/NOT didn't extend across all available filters.

Why it mattered: Four legacy brands already did this in some capacity, and we were migrating their users to Altrata. Parity was the condition of the migration, not a feature request. The cost of missing it was churn and lost renewals.

My position: The immediate ticket scoped boolean to location filters only. I pushed for solving it across all of them, and got the time to do it. A partial fix would have meant rebuilding it later.

PROCESS

The Problem's Problem
Zooming out

Two requirements arrived with almost the same need. **One was urgent** and **small, one larger** and **scheduled for later.** A quick win would hit parity fast, but guarantee a revision loop for everything else. **Double the work, double the time.**

1. Boolean logic in ***Location filters only***
2. Boolean logic in Location filters + ***all other filters***

I made the case for solving it once, properly, and got the scope changed. **Solve for tomorrow, and today solves itself.**
Building piece by piece blurs the end picture. You end up with a crooked face because nobody agreed it was meant to be straight.

Design Approach

I set out a clear approach for my team to proceed with.

Ticket → Define the bigger scope → Research → Design → Test concepts with users → Find out → Build

Raster with AI

Research + Conceptualisation

We started from **past session recordings** of clients using advanced search, alongside **secondary research** into how boolean logic actually works. Boolean had to sit above the launched feature, and designing inside that structure looked like it **would force compromises.**

Rather than assume that, I **split the exploration deliberately.** I directed my designer to work boolean into the existing advanced search, and took the opposite brief myself: question the current approach and **design from a new perspective.** Testing could then tell us whether something better existed.

I drew variations and used **Claude** to translate them into low-fi wireframes.





Inline operators - one flat list, operator lives between rows



Query groups - nested brackets, operator owned by the group



Buckets - drag fields into must / any / exclude columns

Several iterations later, the two briefs had resolved into **genuinely different concepts.** One was a new approach none of the legacy brands had, closer to how comparably complex products work elsewhere but tuned to Altrata. The other embedded boolean into the current structure, with all filters displayed at once.

Concept A (*inline*) **assumes you think in rules, and gives you a blank slate to build one.**



Concept B (*Zara's*) **assumes you think in filters, and fits boolean into the modules you already use.**

The question was what users preferred, how much **complexity they'd tolerate,** and whether one **product could serve both or needed to let users choose.**
Every round of testing ahead was an attempt to answer that.

Design Toolkit
Prepping for Testing

The concepts were still low-fidelity and light on detail, so clients weren't the right audience yet. I proposed **internal product stakeholders as a first check:** people who could tell us quickly whether the two directions worked, aligned with strategy, and were worth taking further.

2	10	20 mins	1
Finalised concepts	Internal Stakeholders (Data, Product, Dev)	per session	week

with interactive prototypes, built with **Claude AI**

Speed as a research method.

Claude turned sketches into low-fi wireframes and interactive prototypes fast enough to test with evidence. That meant **28 participants across 3 rounds in 6 weeks,** clicking real interactions rather than static screens.

For a problem this abstract, that was the difference between opinion and evidence.

Expectation vs Reality
Round 1

Testing returned **a tie.**

we love Concept A! (50%)

"Rule builders are the new age filters now." (Data Product Manager)

"I got something pretty complex in like a minute and a half, that's very helpful." (Account Manager)

"Having and/or idea on each filter makes it quite confusing for me. Whereas this makes it incredibly simple." (Product Director)

we love Concept B! (50%)

"I like to see all the options available to me. That helps me make a decision." (Developer)

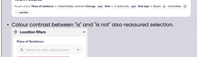
"I feel like it should just be baked in. It's just an extra step that I don't think I need to take." (Senior Product Manager)

"It opens a completely new UI for us to build a query, and it was not very intuitive for me." (Product Manager)

Key highlight.
A visible pattern emerged. Tech-savvy participants preferred the self-build option; less tech-savvy ones preferred it built in. **A preference split rather than a usability issue.**

Insights that fed into the design.

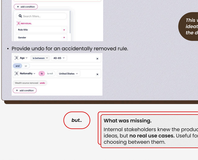
- The live query statement reassured people their filter selections had registered.



- Colour contrast between "is" and "is not" also reassured selection.



- Group multiple values of the same field together rather than stacking them separately. *The UI automatically combine values of duplicate fields.*
- Reduce the burden of knowing what sits behind each filter when the list isn't laid out. Searchable filters.



This was more like group ideation, and it gave us the detailing we needed!

- Provide undo for an accidentally removed rule.



but..

What was missing.
Internal stakeholders knew the product and gave us good enhancement ideas, but **no real use cases.** Useful for shaping the concepts, not for choosing between them.

The Upgrade
Round 2 of testing

To fill that gap I proposed a closer proxy: Commercial and Customer Success, who spend the most time with clients. It also bought time to deepen the concepts before reaching real users.

2	8	30-40 mins	1
Updated concepts	Commercial & CS	per session	week

with interactive prototypes, built with **Claude AI**

Round 2 gave us a lot of insight, alongside a pattern of bias. Close as they were to clients, their read was still secondhand.

Concept A (40% vote)

"wow, this is like super advanced. None of the other logging platforms had this." (Client Success Manager)

"I think it's more clear like what you're adding and what gives you a little more power and control." (Senior Account Executive)

"For me, I like the A as it's been doing this for 14 years, but my clients might get intimidated." (Director of Client Success, Stonnington)

"I worry for the ones who are not tech savvy, this might give them more complications." (Senior Client Success Executive)

"This what... it's sort of forcing you into an order... I think a lot of people build by grouping." (Client Success Manager)

Concept B (60% vote)

"The summary part, to B is just GOLD!" (Client Success Associate)

"I really like that summaries let me able to tell the story myself and think out loud." (Director of Client Success)

"The scrolling up and down was confusing after a point." (Senior Account Executive)

light & simple

heavy & advanced

Most loved
Drawbacks

Key highlight & what AI couldn't see.
Five of eight participants, all experienced at training clients, first tried to type their **whole query into a single keyword field** using parentheses, quotation marks and AND/OR/NOT directly. Muscle memory, acted out silently. No transcript captures that. So I directed the team to run a **manual pass on interactions, facial reactions, and physical movement** toward the screen.





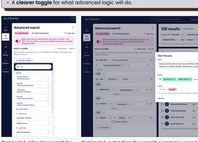
Director of Client Success leaning in. The kind of signal no transcript captures.



Insights that fed into the design.

- On Concept A, most users searched by value rather than by field - **"CEO" rather than "Role title."** A mental model worth designing to.
- Make **keyword search boolean-capable in its own right,** since that's where people instinctively went first.
- The **Concept B** pattern won extended **its exports.** so gift officers and CSMs could keep using what's still see the criteria.
- A **cleaner toggle** for what advanced logic will do.

Sharpened both concepts considerably.



Concept A, allowing search by values while adding conditions.

Concept A, extending the search summary export into sharing an ongoing search with teammates.

but...

Growing pattern.

- **Technical confidence** shaped preference here too, but with more to it. Half the participants wanted Concept A for themselves and Concept B for their clients, assuming from past experience that A would need more hand-holding. That judgment was about people who weren't in the room.
- Concept A was more targeted, built for power boolean users. It's summary was great, but **scrolling through filters made it easy to get lost,** the same problem clients already had with the existing list.
- **Almost everyone wanted parts of both,** neither one whole. That matched our instinct, though it needed real users to confirm.

and...

What should be improved.

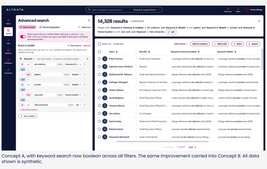
- We showed Concept A first to some participants and B first to others, and saw that whichever came first gave people time to **warm to boolean before the second.** We needed to **reduce that influence** in the next round.
- We asked people to think of a complex query live in the session, which was difficult to do on the spot. A heads-up to **prepare one beforehand** would have helped.

Getting to the clients
Round 3 - A/B Split

To **tackle concept bias, I proposed an A/B split** for round 3, tracking discoverability, onboarding and feature adoption unaided rather than hand-held.
In my experience across enterprise products, **client time is the scarcest resource.** What comes back from commercial is consistently "clients are busy," "we can't bother them now," "show them the final version." Round 3 took three weeks against one for the earlier rounds, and that's why.

2	10	A/B Split	30-40 mins	3
Refined high-fidelity concepts	Clients from a minimum 4 verticals	Design research method	per session	weeks

with interactive prototypes, built with **Claude AI**



Concept A, with keyword search now boolean across all filters. The same improvement carried into Concept B. All data shown is synthetic.

We are at:

- **Round 3 is running now:**
 A/B split with 10 clients across four verticals, tracking discoverability, onboarding and feature adoption.
- **What I'm betting on is Concept C:**
 A balanced in between, a powerful keyword search, and a readable summary of the criteria, letting users move from ready-made filters toward personalised ones at their own pace. The split test exists to check that bet, not confirm it.

...further testings as it demands.